Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 2nd Semester B.Sc. (Hons.) Final Examination, 2020

Course Title: Digital Logic Design
Time: 3 Hours

Course No: **CSE 253**
Full Marks: **60**

READ THE QUESTION CAREFULLY.
There are five sections. Each section carries equal marks.
Figures in the right margin indicate marks.

1) **SECTION-1**

Section No. 1 is based on ***CO1***. There are seven questions.
You have to answer ***ANY 06 (SIX)* out of 07 (SEVEN) questions** from this section.

a) State some major advantages of digital systems over analog systems. 2

b) Draw a pulse train used in digital system and also show the rise time, fall time and pulse width. 2

c) Mention the differences between combinational and sequential circuits? 2

d) Define register. Categorize different types of shift registers with necessary diagrams. 2

e) Give the concept of race around condition and outline its elimination procedure. 2

f) Define binary digits, logical levels and digital waveforms with their necessary diagram. 2

g) Draw the internal structure of a 16×8 RAM. 2

2) **SECTION-2**

Section No. 2 is based on ***CO2***. There are four questions.
You have to answer **any three questions out of four** from this section.

a) Explain logic gate. Draw the symbol and the truth table for 2-input AND gate. Also show the output assuming two arbitrary input pulse trains. 4

b) Illustrate the procedures for implementing the following expressions using appropriate duel symbols:
- X=(A+B)(C+D) 2
- X= AB+CD 2

c) Distinguish between half-adder and a full-adder. Demonstrate the construction of a 16-bit parallel binary adder using 4-bit binary adder ICs with necessary diagram. 4

d) Convert the gray code 1010111 to binary along with necessary circuit diagram. 4

3) **SECTION-3**

Section No. 3 is based on ***CO3***. There are four questions.
You have to answer **any three questions out of four** from this section.

a) Simplify the expressions using Boolean algebra:
i) [AB(C+(BD)′)+(AB)′]C 2
ii) ABC[(AB+C′)(BC+AC)] 2

b) Apply 4:1 and 8:1 multiplexer block in order to implement the following Boolean expression: 4
$\bar{A}BC + \bar{A}\bar{B}C + A\bar{B}C + ABC + AB\bar{C}$

c) A hamming code sequence 1100110 is transmitted and due to error in one position is received as 1110110. Identify the position of the error bit. 4

d) Explain serial-to parallel conversion of data using shift registers. 4

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SECTION-4

Section No. 4 is based on ***CO4***. There are four questions.

You have to answer **any three questions out of four** from this section.

4) a) i) Figure out the need for binary digits in storing and manipulating data in computers. Point out the reasons for using other number systems then. 2

ii) Calculate the number of bits required to represent the decimal numbers in the range from 0 to 999 using: 1

- Straight binary code 1
- BCD code 4

b) Illustrate the minimization of the following expression using Karnaugh map along with minimized circuit diagram.

$Y = \overline{A}BCD + \overline{A}\overline{B}CD + A\overline{B}C\overline{D} + A\overline{B}\overline{C}D + A\overline{B}\overline{C}\overline{D} + ABC\overline{D} + AB\overline{C}\overline{D} + ABCD$ 4

c) Construct the equivalent SPOS and minimum form of POS expression from the following SOP expression using K-Map.

$\overline{A}BCD + AB\overline{C} + ABCD + A\overline{B}C\overline{D} + \overline{A}\overline{B}\overline{C}D + \overline{A}B\overline{C}D + AB\overline{C}\overline{D}$ 4

d) A certain memory has a capacity of 16K×32.

i) Calculate the number of memory words that can be stored. 1

ii) Identify the number of bits per words. 2

iii) Figure out the different addresses required by the memory. 1

SECTION-5

Section No. 5 is based on ***CO5***. There are three questions.

You have to answer **any two questions out of three** from this section.

5) a) i) Justify the statement "NAND gate is known as universal gate". 2

ii) Develop a logic circuit with inputs P, Q, and R so that output is HIGH whenever P is 0 or whenever Q=R=1, using NAND gates only. 4

b) Explain a 4-bit asynchronous binary counter with timing diagram. If each flip-flop has a propagation delay of 10ns, determine the total propagation delay time and maximum clock frequency at which the counter can be operated. 6

c) i) Water is used for manufacturing process in a factory. The water is stored in four different tanks. A level sensor in each tank produces a HIGH signal when the level of water in the tank drops below a specified point. Design a circuit that monitors the water level in each tank and indicates when the level in any three of the tanks drops below the specified point. 3

ii) Develop 6-bit Johnson counter with truth table and necessary circuit diagram. 3

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Good Luck !!!!

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Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc (Honours) Final Examination 2019

Course Title: **Digital Logic Design** Course Code: **CSE-205**

Time: **3 Hours.** Full Marks: **60**

Answer any 6(**Six**) from the following 8(**Eight**) questions. All questions are of equal marks. Figures in the right margin indicate marks.

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1. Give short answers of the following questions: 10
   i) Express the decimal number 158.58 as a sum of values of each digit.
   ii) Subtract -8 from 13 in binary arithmetic.
   iii) Express the binary number 10101.101 as a sum of values of each digit.
   iv) Express the octal number 237.5 as a sum of values of each digit.
   v) Express hexadecimal number 9A5.8 as a sum of values of each digit.
   vi) Add -8 and 17 in binary arithmetic.
   vii) Convert the following binary numbers into octal and hexadecimal form 10011011011.
   viii) Convert the following binary number 110011 into BCD form.
   ix) Convert the following gray code into binary number. 10101110
   x) Determine the decimal value of the signed binary number expressed in 2's complement form: 101110010

2. a) Prove that, i) $A + A'B = A + B$ ii) $(A + B)(A + C) = A + BC$ 2
   b) Design and implement a BCD adder using addition of the following two BCD numbers: 0111, 1001 3
   c) Define fan in with example. Why gray code is not suitable for arithmetic operation? 2
   d) Name and define the Boolean laws with necessary expression and circuit diagram. 3

3. a) Design a 2- bit comparator (truth table, minimum Boolean expression and circuit diagram). Write the expressions for a 3-bit comparator. 6
   b) How multiplexers work? Draw the circuit diagram with waveforms for a 4-input multiplexer. 4

4. a) Convert the following POS expression to an equivalent SOP expression. 3
   $(A + B' + C)(A' + B + C' + D)(A + B' + C' + D)(B + C' + D')(A' + B' + D')$
   $(A' + B' + C + D)\ (A + B + C' + D)\ (A + B + C + D')$
   b) Draw the circuit diagram of a 16x8 RAM using 8x4 RAM chip. 3
   c) Using Karnaugh map convert the given SSOP expression into minimum SOP expression, a SPOS expression, and a minimum POS expression. 4
   $A'B'C'D + AB'C'D + AB'C'D' + A'BC'D + AB'CD' + A'BCD + ABCD' + A'B'C$
   $A'BC'D' + ABCD$

5. a) What is a flip-flop? Define J-K and T flip-flops using block, circuit and timing diagrams. 5
   b) Implement the following Boolean expression using 4:1 and 8:1 multiplexer blocks. 5

6. a) What are the major purposes of using three-state buffer? 2
   b) What is the advantage of R-2R Ladder DACs over binary weighted input DAC? 3
   c) Define the terms of a DAC: i) Resolution and ii) Settling time 3
   d) Write down the differences between combinational circuit and sequential circuit. 2

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7. a) Implement a 4:16 decoder using 2:4 and 3:8 decoder blocks.
   b) Draw a circuit diagram of a MOD-9 synchronous counter with truth table. 5

8. a) Define a register with block diagram. Show the basic data movement in a shift register. 4
   b) Draw a 4-bit bidirectional shift register and explain the working principle with necessary timing diagram. 6

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JAHANGIRNAGAR UNIVERSITY
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc.(Hons.) Final Examination 2018

Course Code: *CSE-205* Title: *Digital Logic Design*
Full Marks: *60* Time: *3 Hours*

Answer any **six** from the following questions.

1. a) Define binary digits, logical levels and digital waveforms with necessary diagram. 3
   b) Name and define the Boolean laws with necessary expression and circuit diagram. 4
   c) Name the universal gates. Why they are called so? Explain with example using one of the universal gates. 3

2. a) Water is used for manufacturing process in a factory. The water is stored in four different tanks. A level sensor in each tank produces a HIGH signal when the level of water in the tank drops below a specified point. Design a circuit that monitors the water level in each tank and indicates when the level in any three of the tanks drops below the specified point. 3
   b) Apply DeMorgan's theorem: 3
   i) $\overline{A + B\overline{C} + D(E + \overline{F})}$ ii) $\overline{ABC + D + F}$
   c) Convert the gray code 1010111 to binary. 1
   d) Show how the following expressions can be implemented using appropriate duel symbols: $X = (A + B)(C + D)$; $X = AB + CD$. 3

3. a) Using Boolean algebra simplify the expression: 3
   $$\overline{A}BC + A\overline{B}\,\overline{C} + \overline{A}\,\overline{B}\,\overline{C} + A\overline{B}C + ABC$$
   b) Covert the SOP expression into SPOS expression: $AB + \overline{A}B + AB\overline{C} + ABC$ 2
   c) Using Karnaugh map convert the SSOP expression into minimum SOP expression, a SPOS expression, and a minimum POS expression. 5
   $$\overline{A}\,\overline{B}\,\overline{C}D + A\overline{B}\,\overline{C}D + A\overline{B}\,\overline{C}\,\overline{D} + \overline{A}B\overline{C}D + A\overline{B}C\overline{D} + \overline{A}BCD + ABC\overline{D} + \overline{A}\,\overline{B}C\overline{D} + \overline{A}B\overline{C}\,\overline{D} + ABCD$$

4. a) Suppose a computer is connected with five input/output devices. How a decoder can select one of the input/output devices to communicate with the computer to transfer data. Explain with necessary diagram. 3
   b) Implement the following expression using 4:1 multiplexer: 2
   $$A\overline{B}C + \overline{A}B\overline{C} + \overline{A}BC + \overline{A}\,\overline{B}C + \overline{A}\,\overline{B}\,\overline{C}$$

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c) Design a 1:4 demultiplexer. 2

d) Draw a 16:1 MUX using 4:1 MUX. 3

5. a) Draw a circuit diagram of a 3-bit up-down synchronous counter with timing diagram for the following up/down control signal: 111100000111. 5

b) Differentiate synchronous and asynchronous counters. Draw a circuit diagram of a MOD-10 asynchronous counter with truth table and timing diagram. 5

6. a) Draw a circuit diagram of a 4-bit parallel-in-serial-out shift register with timing diagram. 4

b) Draw the circuit diagram of a 6-bit Johnson counter with truth table. 4

c) Give an example of an elementary comparator circuit. 2

7. a) How J-K flip-flop can be used as frequency divider? Give example with three flip-flops. 4

b) Draw a circuit diagram of a MOD-8 synchronous counter with truth table. 4

c) Explain the working principles of J-K and D flip-flops. 2

8. a) How memory works? Draw the internal structure of memory with volume 16x8. 5

b) Draw the circuit diagram of a 16x8 RAM using 8x4 RAM chip. 5

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JAHANGIRNAGAR UNIVERSITY
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Hons.) Final Examination, 2017

Course ID: CSE-205
Time: 3 hrs.

Title: Digital Logic Design
Full Marks: 60

Answer any ***SIX*** questions.

1. 4 + 2 + 4
   a) Define binary digits, logical levels, and digital waveforms using necessary diagrams.
   b) Prove that i) $A + A'B = A + B$ ii) $(A+B)(A+C)=A+BC$
   c) Define the Boolean laws with necessary expression and circuit diagram.

2. 4 + 2.5 + 3.5
   a) Apply DeMorgan's theorem i) $\overline{\overline{A+B\overline{C}+D(E+\overline{F})}}$ ii) $\overline{\overline{ABC}+D+F}$
   b) Using Boolean algebra simplify the expression: $[AB(C+(BD)')+(AB)']C$
   c) Show that the following expressions can be implemented using NAND gates only:
      X=(A+B)(C+D); X= AB+CD

3. 6 + 4
   a) Design a 2-bit comparator (truth table, minimum Boolean expression, and circuit diagram). Write the expressions for a 3-bit comparator.
   b) How does a demultiplexer work? Draw the circuit diagram with waveforms for a 4-output demultiplexer.

4. 2 + 3 + 5
   a) Convert the following POS expression to an equivalent SOP expression using truth table:
      $(\overline{A}+B+C)(\overline{A}+\overline{B}+C)(A+\overline{B}+\overline{C})(A+B+C)$
   b) Minimize the following expression using Karnaugh map and draw the circuit diagram after minimization:
      $Y = \overline{A}BCD + \overline{A}\overline{B}CD + A\overline{B}C\overline{D} + A\overline{B}\overline{C}D + A\overline{B}\overline{C}\overline{D} + AB\overline{C}\overline{D} + AB\overline{C}\overline{D} + ABCD$
   c) Map the following SOP expression on a Karnaugh map and find the equivalent SPOS and minimum form of POS expressions:
      $\overline{A}BCD + AB\overline{C} + ABCD + A\overline{B}C\overline{D} + \overline{A}\overline{B}\overline{C}D + \overline{A}B\overline{C}D + AB\overline{C}\overline{D}$

5. 5 + 5
   a) What is a flip-flop? Define S-R and D flip-flops using block, circuit and timing diagrams.
   b) Implement the following Boolean expression using 4:1 and 8:1 multiplexer block:
      $\overline{A}BC + \overline{A}\overline{B}C + A\overline{B}C + ABC + AB\overline{C}$

6. 4 + 6
   a) "J-K flip-flops are used as frequency divider" – Justify the statement with an example (divide by four) with timing diagram.
   b) Draw a 4-bit bidirectional shift register and explain its working principle with necessary timing diagram.

7. 5 + 5
   a) Implement a 4:16 decoder using 2:4 and 3:8 decoder blocks.
   b) Draw a circuit diagram of a MOD-11 synchronous counter with truth table.

8. 4 + 6
   a) Draw the circuit diagram of a 32x8 RAM using 8x4 RAM chip.
   b) Explain the working principle of an ADC with diagram. How does a digital ramp ADC work?

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